STUDENT TEAM WEEKLY REPORT

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Backend Logic Layer, Quiz Generator & Auto-Grading Engine]	**TEAM ID:** 2 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 09/08/2026 **STUDENT LEADER:** Sumegh Bhamre

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• **Core Logic Services:** Implemented QuestionBankService, QuestionService, and QuestionOptionService managing bank lifecycles (DRAFT → SUBMITTED → APPROVED / REJECTED → LIVE). • **Fisher-Yates Quiz Randomizer:** Developed O(N) unbiased random exam paper creation in QuizServiceImpl filtering questions by Subject, Topic, and BTL level. • **Instant Auto-Grading Engine:** Built QuizAttemptServiceImpl using O(1) Hash Table lookups for automatic MCQ grading and score tracking. • **Moderator Review Workflow:** Enabled inline stem, BTL, mark, and option key editing during review before approval. • **Evidence & Repo:** Codebase sanitized and pushed to GitHub branch *logic_layer_sumegh*.	• **Invalid MCQ Options:** Risk of zero/multiple correct choices. *Action:* Built QuestionOptionValidator enforcing 4 choices & 1 correct choice. **GREEN (RESOLVED)** • **Moderation Lock Conflicts:** Typos caused total bank rejections. *Action:* Allowed edits in SUBMITTED state while locking APPROVED banks. **GREEN (RESOLVED)** • **Duplicate Exam Attempts:** Risk of re-taking exams. *Action:* Added attempt locking (COMPLETED state) and SQL ON CONFLICT DO UPDATE. **GREEN (RESOLVED)**	**Technical Guidance Request:** Could the integration team check whether we have written logic's for everything that is going to be there on frontend?

INTEGRATION AND COLLABORATION

- **Interfaces Used:** Connected Java DAOs to PostgreSQL schema (Database.sql).
- **Teams Consulted:**
  - Collaborated with Database team for userId, user role (TEACHER, MODERATOR, STUDENT), and 2-column subject binding.
  - Database team consulted to review the schema and discuss if there are any changes.
  - Integration team consulted to finalize the branch creation, so that they can work efficiently.
  - Front end team consulted for what buttons they would be adding.
- **Pipeline Learnings:** Gained end-to-end data pipeline insights linking Database → DAO Service → REST Controller → Frontend UI.

COMMON KPI EVIDENCES

- **Foundational Behavior:** Quiz generation execution latency < 15ms (O(N) Fisher-Yates execution). Low memory footprint using connection pooling.
- **Continuous Delivery:** Clean Git commit history on branch *logic_layer_sumegh*. Sanitized codebase free of secrets or binaries.
- **Automation:** 100% Automation in Question bank creation and Moderator review.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• **Sumegh Bhamre:** Support Frontend Team with the exploration of more designs to enhance user experience. *(Due: Aug 12, 2026)* • **Sumegh Bhamre:** Execute end-to-end integration testing with Frontend UI & Auth modules. *(Due: Aug 14, 2026)*	